WEEK 8 · SOCIAL REPRODUCTION

CLASS 2

The Labor of Daily Life

READING 1

"Chapter 1: A Child's Day in Howa" and excerpt from "Chapter 2: The Political Economy and Ecology of Howa Village" in *Growing Up Global*
– Katz

Growing Up Global: Economic Restructuring and Children's Everyday Lives | Katz

Week 8, Class 2

Chapter 1: A Child's Day In Howa — pp. 3–22
Chapter 2: The Political Economy and Ecology of Howa Village — pp. 32–39

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1. A Child's Day in Howa

Morning

An Errand. Almost every morning as the sun reached the horizon Ismail awoke, slipped into his flip-flops, and trundled off to the butcher shop to get meat for his family. On most days mutton or goat was available from at least one of the two butchers who worked in the village. Demand almost always exceeded supply, and unless it was one of the three or so days a month when a cow was slaughtered, Ismail needed to be early to get any meat at all.

Although located on different sides of Howa, neither butcher shop was more than a few minutes' walk from Ismail's house. It seemed that slight proximity and family preference combined to favor the butchery of Abdel Monim, and unless passersby advised otherwise, Ismail checked there for meat first. By the time he reached the wood-and-grass shed that served as the butcher shop, Ismail was likely to find a jostling crowd pressed up against its open side. He pushed through men waiting to be served and joined other children, who, having done the same, stood up front, arms outstretched holding money and papers or empty food packages to receive the meat, shouting to Abdel Monim to attract his attention. The butcher chopped the carcass with a knife and an ax, weighing out portions on a balance using brass weights.

At ten years old, Ismail seemed fascinated with trucks, and often propelled himself around the village as if he were one himself. After securing meat for his family, Ismail was likely to make the sound of an engine revving and charge back up the narrow dirt road to the combination of mud and mud-brick walls, thorn branches, and buildings that defined the perimeter of the *hosh,* or houseyard, of his extended family. Most extended families in Howa were patrilocal, and the household of Ismail and his relatively small family of five was located within the hosh of his paternal grandfather.

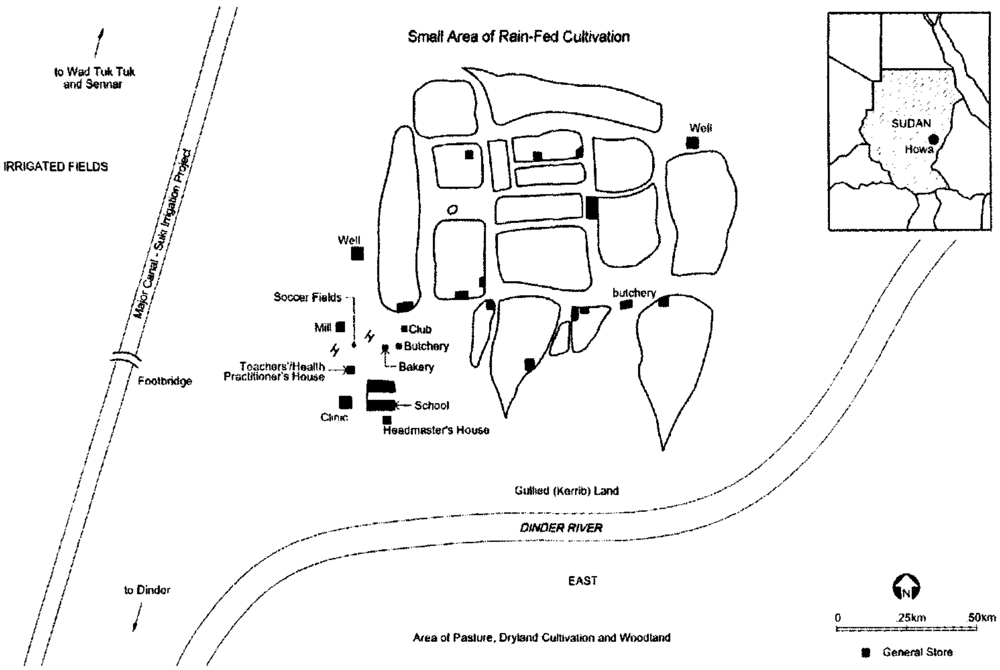

Figure 1. Sketch map of Howa Village, showing clusters of settlements separated by paths, main common structures, and surrounding features.

From the end of 1980 through most of 1981 I lived in this hosh—six months of the time with my then-partner, Mark—in a mud house that had been abandoned recently by a household moving to a newly constructed brick house. Each morning Ismail bought meat for us as well. We heard his "engine" above the sound of braying donkeys and bleating sheep a few moments before he peeped around the rag rug and thorn branches that served as a door to our grass enclosed "veranda." He gave me a handful of goat meat wrapped in the last pages of a *Time* magazine bought months ago. His pleasure was apparent as he unwrapped the paper and showed us a relatively unstringy and boneless quarter kilo.

Off to School. Sometimes Ismail stayed to have tea and cookies, but on this day he declined; having been long at the butcher shop he did not want to be late for school. He picked up the package of meat for his family and crossed the small yard between our houses. At home he found some sweet milky tea his older sister had prepared for him waiting on the embers of their charcoal stove. He gulped down the tea, slipped his blue ankle-length cotton *jallabiya* (caftan) over his coarser knee-length muslin one, grabbed his cloth book bag and raced back down the road to the village elementary school, where he was in second grade.

From the school yard one could see the butcher's shed, a few dry goods stores, and the mud oven of the bakery, which produced bread during only part of the year. A road formed between these mud-and-grass buildings and led out past the gristmill, through some acacia shrubs to the major canal of the Suki Agricultural Project and the agricultural fields lying on the other side of it. In June, by seven o'clock, when the teachers

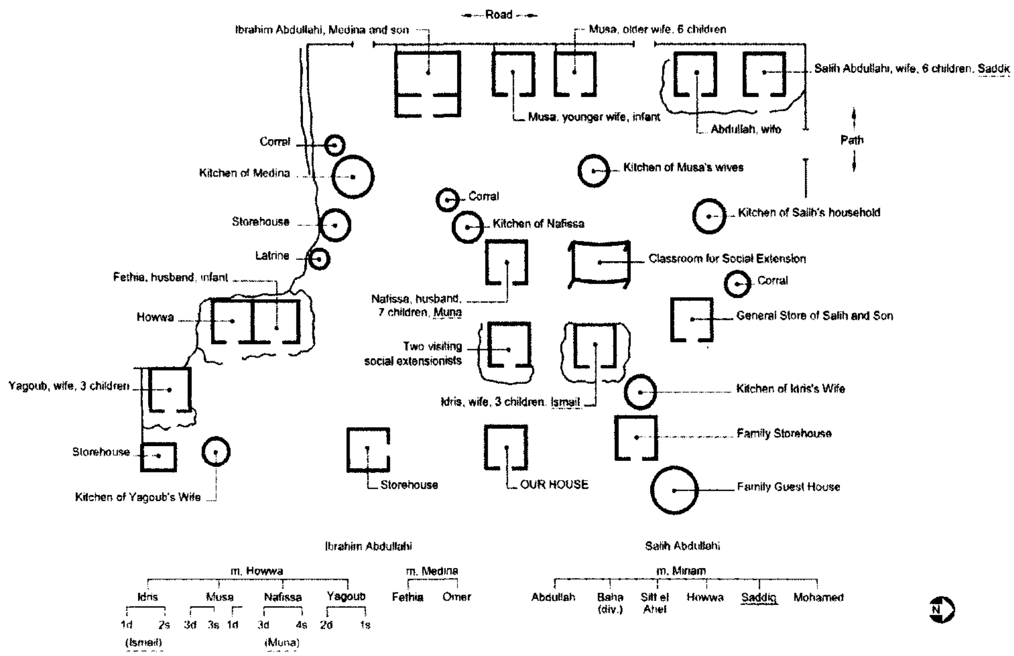

Figure 2. Houseyard (hosh) and kinship relations of Abdullahi family.

began to call the children to attention, most farmers were long gone, clearing their fields and cleaning the irrigation ditches along them in anticipation of the planting season, which began with the rains in late June or early July.

With the hottest months of April and May over and the first rains beginning, the summer solstice brought about a time of renewal during good years in Howa. In a few days school would close, in part to allow the teachers to return to their home districts during the rainy months from July through September, and in part to free students to help their families with the agricultural tasks of this busy season. In the coming weeks, for example, Ismail and most other children nine to twelve years old would spend their mornings in the fields planting groundnuts, sorghum, and cotton in rows drilled by the digging sticks of their fathers and older brothers.

Herding. The ground was still parched, however, and even after a few rains there was only a hint of new green vegetation. June was an especially difficult time for the animal population of Howa. Even dry vegetation of minimal quality was sparse by this time of year, and shepherds had to go out with their flocks from dawn until dusk so that an adequate diet could be maintained. The long days of grazing notwithstanding, milk production had all but stopped and several animals in the village had died of starvation-related causes.

Ismail's eleven-year-old cousin, Saddiq, was the shepherd for his family's flock of approximately fifty goats and sheep.[1] In these weeks of extreme heat, when daily maximum temperatures averaged over 40° Celsius (104°F) (Ferguson n.d., 20), Saddiq's parents often saw to it that he did not herd all day, but rather was relieved for part of the time by his slightly older, but less reliable, brother, Mohamed.

One day when I went shepherding with Saddiq, Mohamed came along. We set off at six-thirty in the morning. Saddiq made Mohamed guide the animals over the footbridge spanning the major canal of the irrigation project bounding the settlement. With a series of special clicking and yelping noises, Saddiq then led the animals to drink briefly from a small canal south of the bridge before continuing southwest to a depression he had heard was green because a broken irrigation ditch had flooded the area. The depression or *maya* was indeed green, and Saddiq and Mohamed let the animals graze, joining two friends who had met them along the way to play *shedduck,* a game in which players hop holding one leg behind them, madly attempting to knock down their opponents while remaining standing themselves. The boys played several rounds along with some other active games and then sat in the as-yet unplanted fields talking. Finally they rounded up the animals and led them to an area of *Acacia seyal.* Because of the dearth of green vegetation, Saddiq and the other herdboys scaled some trees and, with axes brought along for this purpose, lopped off branches for fodder, clicking and calling to the animals as they did.

By nine o'clock the sun was high in the sky and the boys were ready for breakfast. One of them brought out an old enamel bowl, another had a wad of sorghum porridge. Saddiq ran from sheep to goat to sheep squeezing a bit of milk from each one. With Mohamed holding the hind legs of the unwilling and rather dry animals, Saddiq

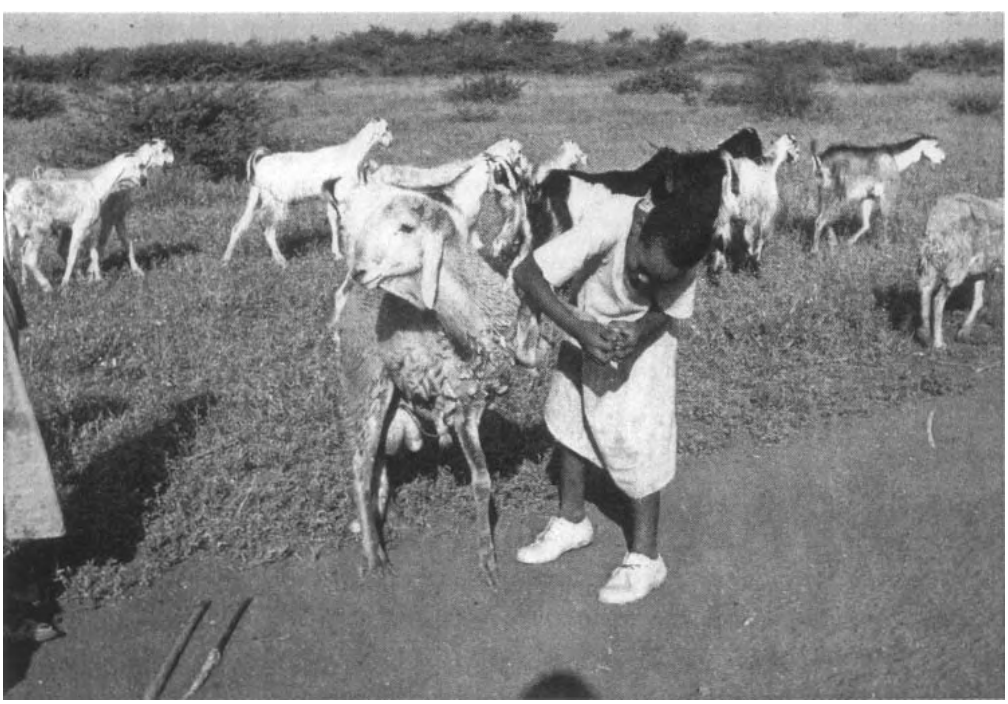

Figure 3. Shepherds not only were attentive to watering their flocks and ensuring that they circulated through various pastures (at once a means of maintaining grazing areas and diversifying their animals' diet) but also were sensitive to their overall look—making sure no animal was missing, watching their gait, and checking whether they showed any symptoms of distress. Here a shepherd removes a thorn from the foot of a sheep as they head out of the village along the major canal just after the rains had begun.

managed to get enough milk for the five herdboys then present. The boys added the porridge to the bowl of milk and gathered around it to eat in a bit of shade cast by one of the trees.

Breakfast for Ismail. At about the same time, the schoolchildren were dismissed for breakfast. They spread out quickly along the fan of dirt roads that extended from the school area. Ismail's fifteen-year-old sister, Selwa, had prepared a typical breakfast of sorghum porridge and a soup of reconstituted pulverized dried okra, spices, and water. Ismail and his seven-year-old brother, Ali, who had just started first grade, washed their hands to join their father for breakfast. They squatted around a common bowl to eat in the shade of their grass-covered veranda and followed their meal with hot sweet tea. Selwa and the children's mother ate their breakfast from a separate bowl. The two groups sat near one another and conversation flowed freely between them.

An Errand. Ismail was still drinking his tea when his mother asked him to go get three piasters'[2] worth of green coffee beans, enough for the several small cups of strong spicy coffee that she and her husband, like most adults in Howa, enjoyed after breakfast. He took the money from her and somewhat begrudgingly trudged off to the nearest dry goods store likely to be open. Along the way he stopped to watch some men building a mud house and became so absorbed that only after a few minutes did he remember the coffee and continue on his way. He bought the coffee beans and again stopped to watch the men at work. When some boys passed him on their way back to school, he set off quickly for his house. Screeching to a halt in front of his mother, he dropped the small newspaper cone of coffee beans, grabbed his book bag, and raced back down the road to school.

Domestic Chores. In his haste Ismail barely noticed his ten-year-old cousin, Muna, standing between their houses holding her infant brother. Muna, who envied Ismail for being in school, glared at him as he ran by. Muna was the daughter of Ismail's father's sister and the fourth of seven children. Her thirteen-year-old sister, Ajba, was the oldest girl in the family and shouldered most of the responsibility for helping the children's mother with domestic work, including cooking, child care, getting firewood, and fetching water. Because of Ajba, Muna was not directly responsible for many of the tasks undertaken by girls of her age in Howa. Despite her relative freedom, Muna's parents had not seen fit to send her to school, largely because they did not think school was important for girls and also because they seemed to feel it was not proper for girls to study side by side with boys.

Muna often helped out in the other households within the family compound. She assisted her grandmother and one of her aunts, whose three children were all under six years old, with domestic chores such as sweeping and dishwashing. In her own household, Muna ran errands for her parents, swept, and helped to take care of her two younger brothers, who were four years and eighteen months old. In recent months, as Ajba neared puberty and thus the beginning of restrictions on her mobility associated with *purdah* (the seclusion of women from public observation), Muna had begun to take greater responsibility for some of her sister's tasks in anticipation of the elder girl not leaving the houseyard unnecessarily and probably marrying. Muna helped her older sister and mother wash clothes, occasionally accompanied her grandmother to glean charcoal from abandoned kilns around the village, and was learning to fetch water from the village well, in part by bringing water to her grandmother in gallon-size containers.

Fetching Water. After Ismail passed on his way to school, Muna went to the well to fill a small, plastic jerry-can for her grandmother. Because Ajba had been largely responsible for this task, Muna was not as adept at fetching water as many of her peers. Some girls of ten years rode donkeys to fetch as much as ten gallons of water. With only minimal assistance, they lifted the full five-gallon jerry-cans, each weighing forty pounds, onto the wooden bar that crossed the saddle of their donkeys. Other girls this age were able to carry a full five-gallon vessel on their heads, but most were not yet strong enough. Most of these girls carried one- or two-gallon containers on their heads. Muna, however, was inexperienced and not yet able to balance even a small jerry-can on her head for the two hundred or so meters between the well and her house.

When Muna reached the well she was reticent to move forward into the throng of people that usually surrounded the water taps during the well's limited hours of operation. While Muna's hesitation may have been a result of her inexperience, it also may have been a considered response to the pushing and shoving that often occurred at the well. Others with considerable experience were timid when faced with the crowded water table.

Certainly it was a question of style rather than experience with ten-year-old Sofia, who went to the well frequently. Sofia was the oldest of five children and got water for her family everyday, sometimes several times a day. One day I accompanied her to the well just after it had opened and found it already crowded. Most of the crowd was made up of kids who pushed and shoved one another in an attempt to get water before anyone else. Sofia was quite timid compared with most of the children. She waited several minutes before going near the meter-high table with its four overhanging spigots. In the interim the jockeying and jostling intensified. After several more minutes, during which she frequently looked over with a mixture of bemused helplessness, exasperation, and loathing on her face, she made her move and climbed onto the concrete table that supported the pipe and spigots. On the edge of the wet table Sofia looked a lot smaller than most of the other kids. After another few minutes, she succeeded in getting her bucket under a spigot. When she removed it in order to rinse it out, someone grabbed her spot. With a bit more pushing her foothold became more tenuous. Her balance was thrown off further because she was hanging onto her bucket. With another shove in her direction, Sofia fell off the table and splash-landed sitting upright in the mud puddle that surrounded it. She seemed to have scraped her leg along the edge of the cement table as she fell, and she looked shocked, hurt, and sad when she landed. Sofia's older and tougher female cousin took her bucket and filled it as Sofia wrung her dress and stood crying at the end of the table. She took the full bucket from her cousin, gingerly put it on her head and, still looking a little defeated, walked off with a group of sympathetic friends.

Given this sort of scene, Muna's cautious approach to filling her jerry-can was well founded. She went up to the table during a lull in activity and pushed her plastic jug up to the faucet. She managed to hold her own until it was full, then slogged through the mud and ankle-high water around the well, holding the heavy jerry-can with her extended arm. Once on dry ground, Muna put the jug down, wiped her face with her wet hands, and headed up the slight incline toward home, holding the full container in both hands. Carrying things on the head was a task children learned by doing, and once Muna was up the hill and out of sight of the other girls she put the tightly closed one-gallon Mobil Oil container on her head and attempted to balance it as she walked. She wobbled while walking and more than once had to steady the jug with her hand, but she seemed proud that it never fell, wearing her aching neck around like a badge of glory.

An Errand. After pouring the water in her grandmother's clay water container, Muna ran the short distance back to her house. Her mother was in their thatched round kitchen-house preparing the afternoon meal on the same fire she had used for the morning porridge. When she heard Muna, she called her to come in the kitchen. It took a few seconds for Muna's eyes to adjust from the bright sunlight to the kitchen, which was lit only by the fire and a few small windows in the thatch. As she stood in the doorway her mother gave her two piasters and asked her to buy some garlic to season the okra-based stew. Muna took the money with little enthusiasm and a touch of pique at being asked to do another chore. As she headed across the houseyard to the metal gate next to her grandfather's one-room mud-brick house, her mother called her back. Muna found her digging a few piasters out of the old leather wallet she wore on a leather string around her neck. She handed Muna four more piasters and told her to get a package of cookies as well. This time Muna sang and walked with an occasional skip down to the shops near the schoolyard. When she returned, she took one of the cookies and, without being asked, gave the remaining ones to her younger sister and two younger brothers.

Domestic Chores. Muna found the wooden mortar and heavy metal pestle and began to pound several cloves of garlic to add to the gumbo. Keeping an eye on her youngest brother, she squatted in the shade in front of the house and pounded deftly and swiftly until the garlic was a smooth paste. She gave it to her mother who had finished making the day's *kisra* (sorghum pancakes) and was beginning to fry onions for the gumbo.

Midday

Play. Muna picked up her eighteen-month-old brother and walked over to the adjoining yard of her great uncle—Saddiq's father—to seek out his daughter, nine-year-old Howwa. The two girls, fast friends, spent some time hanging out and talking in the shade of Howwa's family's veranda. After a while they began to gather bits of material, found objects, and homemade straw dolls to play "house." Just as they began to demarcate a place for their play, a teenaged relative of the girls came over holding her baby and told them that her father—Muna's grandfather—wanted help shelling groundnuts to be planted in the coming weeks. In one move the girls dropped their things, Muna picked up her brother, and they sped across the yard to join in the nut shelling.

Shelling Groundnuts. In the two months preceding the planting period, each of the 250 or so local tenants of the state-sponsored agricultural project needed to shell several sacks of the previous year's groundnut crop for seed. Every household accomplished this task with the assistance of child labor. Most children were paid for shelling groundnuts, even within their own extended families. There was no difficulty in finding ready hands for the task because everyone seemed eager for a little pocket money.

When Muna and Howwa reached the house, they found about ten of their cousins and neighbors already at work in the shaded veranda. Muna's grandfather handed them a large bowl brimming with unshelled groundnuts and each girl took another bowl for the shelled nuts. They found a couple of spots on the earthen floor and sat down. Muna plunked her brother down next to her and gave him a handful of shelled nuts to eat. The girls worked quickly, talking, singing, laughing, and eating as the unshelled nuts disappeared, the bowls of shelled nuts filled, and the husks scattered over all exposed surfaces. After a couple of hours, her brother having long since wandered home on his own steam, Muna brought her bowl to her grandfather, who was seated in his vinyl-strung chair at the back of the small veranda. He used a small tea glass to measure the shelled nuts. The going rate was a half-piaster per five-ounce tea glass, and in two hours, Muna had earned three piasters. She immediately went to buy one piaster's worth of candy and then gave the other two piasters to her mother.

Relaxing. Muna returned to her house in the heat of the day and found her mother and three of her aunts sitting in the veranda drinking coffee, talking, and tossing cowrie shells. The women of Howa often passed the time telling fortunes and exchanging information, gossip, and advice based on their interpretations of the patterns made by six cowrie shells tossed in turn by each participant. Certain women were considered expert interpreters and fortune tellers, and it was usually their presence (and their shells) that sparked a session. Muna sat quietly near the women and listened to them talking as she absentmindedly ate a handful of sugar given to her by her mother as she sweetened the coffee.

While the women talked, Muna's father napped on the other side of the veranda. He had spent the morning clearing their tenancy in preparation for planting. When he stirred, the women began to leave and Muna's mother prepared to serve the midday meal.

Dinner. In Howa, men, women, and young children ate separately. With his oldest son studying typing in the town of Sennar, approximately forty kilometers away, and his next oldest son attending junior high school in Dinder, the district capital, about twenty-five kilometers away, Muna's father ate alone. Although Muna sometimes ate with her mother, grandmother, and sister, Ajba, she usually ate with her three younger siblings. Before eating, she helped the two youngest wash their hands and had a minor squabble with her six-year-old sister when she questioned Muna's authority on such matters. The children squatted on the ground to eat from a tray of neatly folded sorghum pancakes and a bowl of gumbo. The three older children pulled off pieces of the pancakes with their right hands and dipped them into the hot gumbo. Muna helped her youngest brother by dipping small pieces of pancake into the stew and putting them into his right hand to eat by himself once they had cooled. Every few days when the family had meat, each child got one or two small chunks of it, and quarrels sometimes erupted over who got more.

Domestic Chores. After the black sugary tea that followed dinner, Muna picked up the short straw broom and, beginning in the back by the wall of the house, swept the whole veranda clean. She picked up the sweepings with an old metal sheet and, because most of it was organic, tossed it to their donkey, who ate most any scraps.

Play. The intense heat of midday precluded strenuous activity. After taking their main meal most adults napped, visited, or took care of more sedentary chores, such as mending, rope making, mat making, and accounting. Almost everyday during this quiet time, ten-year-old Rashid, who had just begun third grade, led his mostly male and slightly younger friends in one of three dramatic-modeling play activities. The content of each game is suggested by its name: *dukan* (store), *hawashaat* (tenancy), and *bildat* (subsistence fields). These games involved the children in elaborate enactments in miniature of the behaviors and tasks associated with each of these settings.

One day I found Rashid in his family's houseyard sitting under a shade tree, just putting the finishing touches on a toy tractor he had made with the help of his fourteen-year-old brother, Adam, who was home on vacation from junior high school. It had been Rashid's idea to build the tractor, and he had completed most of the work on it himself. When I arrived he was pushing it back and forth while Adam made a plow to attach to it.

The tractor, which measured about eleven by eighteen centimeters and just over eleven centimeters high, was a good example of the ingenious motor vehicles fashioned by children in Howa from things that would be considered junk to the untrained eye. The chassis was a piece of wood bridged by two strips of metal that arched across it one-third of the way from each end. The tires were circles cut from the soles of old plastic thongs and turned on axles of thin metal rods. Its headlights were made from the circular nubs that rivet the straps of flip-flops into their soles. The final touch was a plow fashioned from a thin sheet of rusted metal that was attached to the rear of the tractor.

Once the plow was attached, Rashid and two male friends about nine years old, along with two boys about four years old, began to play hawashaat (tenancies). They rolled the tractor to the shady area in front of the men's sitting house and began to play by modeling the dirt into raised squares about twenty-five by forty centimeters. These raised plots were their agricultural fields, and each boy plowed his field with the tractor, occasionally using his fingers to form ridges when the finicky plow became too much of a bother. After the rows were completed they made *teganet,* the impoundment ridges that run across groups of furrows and control the flow of water from the irrigation ditches to the crops. When the ridges were in place, the boys planted groundnuts by

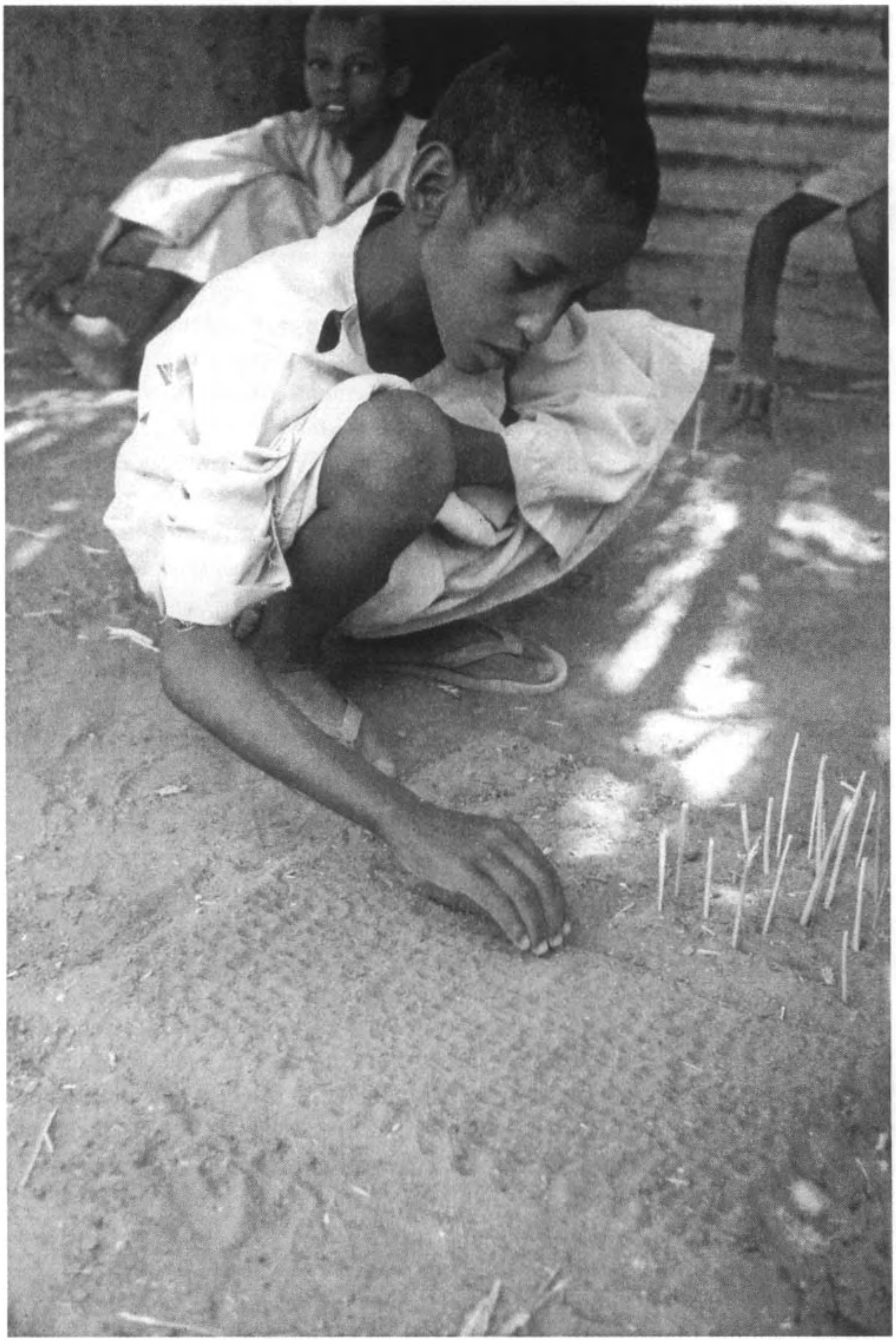

Figure 4. In almost daily games of "hawashaat" and "bildat," Rashid and his friends worked through the agricultural sequence of irrigated agricultural and rain-fed subsistence cultivation, respectively. Playing out the discrete work activities of cultivating various crops, the games also embraced the political-economic and social arrangements associated with the two forms of agriculture.

sticking date pits they had collected and saved for this purpose vertically into the plowed rows. Although sometimes they used limited quantities of real water, on this occasion they "watered" the crop by sprinkling sand on it. The boys were aware of the method of irrigation in the real fields, and seemed to enact this knowledge by watering only in the furrows between the rows, as if the water had flowed there from the canals.

When they finished irrigating, the boys began to weed the fields using miniature short-handled hoes that closely resembled those used in the working fields. Each boy made his own hoe from a short piece of thick straw and a small piece of thin metal broken into the customary wedge shape. When the weeding was completed, the boys harvested the groundnuts by piling the date pits in the center of the field. During the harvest in Howa, the whole groundnut plant is pulled and piled similarly in the center of each field. The children used tomato paste cans as sacks and filled them with the harvested date pits. They loaded the sacks onto the tractor and carted them several meters to the shade tree that was their storehouse.

They repeated this complete agricultural cycle, and then in the third round grew cotton as well, which they represented, perhaps not accidentally, with goat dung. After the cotton harvest Rashid went to each boy's field, counting and weighing the crop in the tomato paste cans. He paid the boys shards of "china money" according to the size of their yields. While the going rate for a sack of groundnuts mirrored the market price during the preceding season, the children's price for a sack of cotton was considerably less than the actual price. This confusion may have been rooted in the obscurity of cotton prices for adults in Howa as well. Tenants did not sell their cotton on the open market as they did their groundnuts. Not only did they not receive direct payment for their product, but they were paid by the Suki Project authorities only after all agricultural, service, and marketing expenses had been deducted. It was difficult for most tenants to determine the precise rate at which they were paid, and it follows that cotton prices may have been less accessible to children as well.

The quiet hot hours of early afternoon were almost all spent by these boys playing several rounds of "tenancies,"or a variation called bildat (rain-fed subsistence fields) in which they reenacted the dryland cultivation of sorghum and sesame that characterized the village before the Suki Project. The boys also played another dramatic-modeling game called "store," where they spent those shards of china on everyday commodities represented by found objects, like bits of metal and glass, battery tops, and scraps of packages. While the village rested, these children carried on its work at a miniature scale—playing with their possible adult roles—using the debris they found around them.

Late Afternoon

Fetching Water for Sale. As the day became cooler, the village began to burst with activity and by four o'clock people were up and about as if it were morning all over again. Since diesel fuel was expensive and not easy to come by, the village well was generally

Figure 5. In playing "dukan" (store), Miriam and other children enacted a riot of consumption that in its intensity far exceeded existing commercial patterns in Howa but certainly anticipated what was to come. "Found objects" (otherwise known as garbage) made up the merchandise, and shards of broken crockery, "china money" (sometimes earned playing "tenancies" and crossing over to games of "store") provided the medium of exchange.

open only a couple of hours in the morning and again in the late afternoon. I often knew that it was time to resume working outdoors when I heard the first child coming up the road hawking water. About fifteen boys and girls between ten and fifteen years old sold water regularly in Howa. Many of these young people had donkeys of their own for this purpose, while others relied on the availability of their families' animals.

In a typical morning or afternoon a youngster selling water made at least one trip for his or her own family and returned to the well four to eight more times to fill a pair of five-gallon jerry-cans and hawk them in the village. Each pair sold for the equivalent of about twelve cents, and children generally contributed their earnings to their households. Indeed, Talal, who was a regular water seller in Howa and the oldest child in his family, had withdrawn from school midyear in part to provide this income for his family and in part to work with his father in the fields and at other tasks.

Making Charcoal. The hour before sunset was also a time when people, young and old, went to collect or cut firewood. Ten-year-old Sami, the middle child of three and the oldest boy in his household, went in search of firewood almost daily in the late afternoon with his elderly but quite energetic father and his thirteen-year-old sister. Sami's father was not a tenant and earned an extremely modest living primarily from the sale of charcoal he produced, lumber he cut, and vegetables he cultivated on the riverbank during the rainy season, supplemented with earnings from his occasional agricultural labor during the season. Until the rains soaked the ground and the Dinder

Figure 6. About 10 percent of boys and girls in the village supplemented their households' income through the sale of water. Drawing water from the wells, the river, and even the irrigation canal, young people like Talal hawked their ten-gallon loads for about a penny a gallon to households that either were too busy or were without child labor to provide their own.

River again flowed, Sami's father produced charcoal almost biweekly from an earth-mound kiln he fashioned beside his house. Each day, in the early morning or late afternoon and sometimes both, he and his children crossed the dry riverbed in search of wood suitable for charcoal production. Because the wood supply around Howa had diminished in recent years, most such wood—which only a few years before would have been considered too small—was at least a half hour's walk away.

On a typical day, Sami's father took along Sami, his daughter, Maha, and a nephew about Sami's age. Any number of little ones might tag along, and on one of the days I accompanied them, two boys and a girl less than six years old joined the procession. From the family's round grass house we followed a deep gully to the riverbed. We maintained a swift pace as we trooped across the sandy river bottom and up its steep eastern bank. On the other side, we headed further east through a large and still bare depression that, once the rains came was an area of sorghum cultivation for a neighboring village, until we came upon a stand of *Acacia seyal.*

Sami's father immediately began chopping down a relatively large tree. Within a few minutes Sami took his cousin's ax and began to chop down an acacia, about seven centimeters in diameter, designated by his father. Although far from being as deft, fast, and strong as his father, Sami made good progress cutting down the tree and was clearly familiar with the work. Before Sami was done chopping, his father sent him, Maha, and their cousin to a nearby area to collect wood they had chopped on a previous day.

After about ten minutes the children returned, loaded with wood. When they did so, Sami's father, who had continued to cut down trees and branches, was ready to head back to the village. They would have left with Sami's half-chopped tree to be finished another day, except that I asked if I could see him finish. He accomplished this swiftly, and then they all left, leaving the newly cut tree to be collected at another time.

On the trip home each of them was loaded to capacity with wood. The girls carried the wood on their heads, while Sami's father and the boys carried their loads on their shoulders and backs. The children bore their heavy loads without complaint for the half-hour walk home. Even the youngest girl, who was no more than four years, carried her large load with grace and assuredness. At home, they added their wood to the pile.

After about seven days of this back and forth, there was enough wood to make a charcoal mound. Sami and Maha participated in this part of the task as well. The children helped their father to pile the wood tepee-style around a mound of dirt and dried grass. After building a large cone of wood, Sami and Maha helped their father cover it with a thick layer of straw. While their father raked up straw near the kiln, the children went further afield to gather handfuls of straw fallen from the grass walls, fences, and roofs near their house. Once the wood was completely covered, their father shoveled a layer of dirt over the whole pile. The children rarely assisted with this part of the task. From a small opening in the dirt at the base of the mound, which by then was at least waist high, Sami's father ignited the grass with a hot coal. Once the straw began to burn, he raked some dirt over the opening and left the wood to smolder for two to three days. The earthen kiln was checked throughout this period to ensure that it continued to burn and that no flames erupted through breaks in the dirt. When the wood was completely carbonized, Sami's father spread it outward with a rake and covered it completely with dirt to stop it from burning further. After the charcoal cooled for a day or so, the children helped their father pile it into large burlap sacks. The sacks were sold within or outside the village for the equivalent of between US$2.50 and $3.75 apiece.

Most families in Howa could not afford to use much charcoal, and relied almost exclusively on wood for cooking. The labor of children was integral to procuring a sufficient wood supply for most households. Nine-year-old Awatif, the youngest child in a household with three other daughters twelve to sixteen years old and a son of thirteen, generally collected firewood with her friends a couple of times each week. Awatif was one of the few girls in Howa who attended school, and therefore could only collect wood in the late afternoon and on days off from school.[3]

Play. Much of the time, Awatif's twelve-year-old sister, who did much more household work than she did, collected the fuelwood supply. Thanks to her sister's work, Awatif was often free to play dolls and "house" with her friends. On many days when I visited in the late afternoon, I found these girls playing with dolls in the dramatic-modeling game of "house."

One day I found Awatif and two friends playing along the wall of the veranda with dolls they had made of two sturdy pieces of straw crossed so that the horizontal piece served as the arms, and the vertical piece the head to toe. Each doll had a name

and was dressed in scraps of fabric. The dolls were male and female and of all ages. The girls manipulated them in and around houses that they established with dividers made of shoes, mortars, pestles, bricks, and pieces of tin cans. They used all manner of found objects for props, including an enamel cup, a knife fashioned out of a tin can, shards of glass and crockery, can lids, battery tops, small bottles, grass, razor-blade wrappers, a soap carton, a hollowed-out D battery, empty food tins, dirt, charcoal bits, and cardboard.

Awatif and her friends used these props to play out a range of domestic activities. As I watched, one girl had her doll prepare sorghum pancakes and stew by spreading dirt on a can top, then breaking up pieces of grass and leaves and putting them in a shard of glass. In the course of the afternoon the dolls cooked, cleaned, ate, went to the well, ran errands, and visited. The girls clearly enjoyed this game, in which they experimented with both their present tasks and potential adult roles. They played "house," or a permutation of it without dolls, at least four times a week. They stored all their treasures carefully in boxes and added new found objects as they came upon them.

As I walked home from Awatif's house, I passed by the schoolyard where the teachers, the village medical practitioner, and several other young men were playing soccer. Nearby, and inspired by their example, several boys between nine and fourteen years old played soccer using a wad of socks and rags as their ball. During the relative cool of the hour before darkness, the streets of Howa usually filled with regular "sock soccer" games. Most boys ten and older participated at least occasionally in these localized games.

Evening

The sun was setting as I made my way home, but since it was almost a full moon, night merged with day rather than snuffing it out. The full moon is a special time in villages without electricity, and all of Howa seemed to be up and moving around in the moonlight. Whenever possible, people celebrated weddings and circumcisions during the full or waxing moon. Even without a special occasion, children danced, sang, and played well into the evening bathed in the moon's bright light. On one of these nights we stood quietly outside our house and heard makeshift jerry-can drums played by girls and boys throughout the village. After several glasses of tea we took a walk and came upon about ten boys in the street singing loudly and accompanying themselves by drumming on a jerry-can and clapping. Elsewhere, we found about fifteen girls in a household compound singing, drumming, and taking turns dancing. I danced one long song with Sofia and a friend of hers. It was fun and felt almost magical to establish the old and familiar communion of dancing with the children.

A Child's Day in Larger Perspective

The acquisition and use of environmental knowledge are key aspects of children's socialization. These practices are particularly important in agricultural settings, where environmental knowledge is basic to the productive use of resources and where children's labor figures centrally. Enduring production pivots on the social acquisition of such knowledge as a critical aspect of social reproduction. Political-economic and political-ecologic change of the sort affecting Howa after its incorporation in the Suki Agricultural Project can disrupt these material social practices, as well as the social relations that bind production and reproduction. Acquired knowledge may no longer be apposite, bodies of knowledge may be altered and recomposed more intensely than usual, modes of sharing knowledge may be displaced, and sources of knowledge may be discredited.

This book examines the production and exchange of environmental knowledge in Howa following its inclusion in a state-sponsored agricultural development project geared to production for the world market, and the play of these material social practices over a number of years as the children who are its protagonists and focus came of age. The content, acquisition, and use of environmental knowledge are addressed as mutable cultural forms and practices, structured and shared within a particular socioeconomic matrix and inseparable from the labor process and its underlying relations of production. My aim was to discover and illuminate not only the nature and structure of these cultural forms and practices and their imbrication with political-economic and environmental changes in Howa, but also their meanings for children and their families. By locating this study in a rapidly changing society, I hoped to develop a more dynamic understanding of the relationship between a particular material social practice—the production and exchange of knowledge—and the evolution of the relations and processes of social reproduction. My underlying concern was with resistance and persistence in everyday life.

This project has been long in the making and my initial approach to it grew out of the social theories of Marxism and feminism, and my frustration with the way most Marxist work at the time either subsumed social reproduction to production or imagined it solely as a function of the state, and thus failed to theorize or interrogate it adequately (see Dalla Costa 1972 for a notable and compelling exception). With the obvious exception of the work associated with feminist theory, as well as with the Birmingham Centre for Critical Cultural Analysis and the Social History Workshop, the critical practices of everyday life—wherein social relations were enacted, maintained, and altered—were either invisible, assumed to be unproblematic, or presented in a functionalist manner in most analyses of social life or the operations of political economy.[4] The concern with social reproduction among those scholars who did address it was commonly intertwined with a search for resistance, as was the case for me. But as interest in resistance blossomed in various quarters, its definition became increasingly amorphous and the measure of its effect ever more slight. The attenuation of resistance was compounded by the ascendance of poststructuralism during the 1980s and 1990s as many authors appeared to suggest that the material grounds of production no longer mattered. Yet in the critical practices of everyday life—among other aspects of social reproduction—the conditions of production are ensured. If the terrain of their possibility is profoundly material, so, too, are their operations and effects. One of my aims, then, is not only to examine the workings of one aspect of social reproduction closely, but in the process to problematize the very notion of resistance, defining it more rigorously and differentiating its various forms analytically and practically. This project is taken up in chapter 9.

As suggested in the preface, social reproduction encompasses that broad range of practices and social relations that maintain and reproduce particular relations of production along with the material social grounds on which they take place. Social reproduction has political-economic, cultural, and environmental aspects, each of which bears on the geographies of children's everyday lives. The political-economic aspect of social reproduction encompasses, for example, the reproduction of work knowledge and skills, the practices that maintain and reinforce class and other categories of difference, and the learning that inculcates what Pierre Bourdieu refers to as the habitus, a set of cultural forms and practices that works to reinforce and naturalize the dominant social relations of production and reproduction (Bourdieu 1977; Bourdieu and Passeron 1977). If the political-economic aspect of social reproduction takes place in some combination of the household, the market, the workplace, "civil society," and the state, the relationship and balance among settings varies across time, space, class, and nation, among other things. The gender division of labor within the household, which is itself historically and geographically contingent, commonly presumes women's responsibility for most of the work of reproduction, including, for instance, child rearing, food collection and preparation, cleaning, laundering, and other tasks of "homemaking." With wealth and "development," an increasing number of these tasks may be provided through the market and purchased, depending on household circumstances and other socioeconomic factors. Prepared foods, domestic assistance, child-care services, and the like may lessen household work for some and "free" some women's time for participation in the paid labor force or other activities, but they do not alter the gender division of labor or the social relations of production and reproduction that undergird and are sustained by it. Likewise, the state has historically played a role in social reproduction. From state-subsidized electrification, water supplies, and sewage treatment, to schools and health-care services, as well as a variety of goods and services associated with the welfare state, the state has long been implicated in social reproduction. The varying role of the state across history and geography also affects the balance among the various constituencies in how social reproduction gets carried out. While in some instances, certain aspects of social reproduction, such as child care or meals, are provided through the workplace or through organizations associated with "civil society," it remains that women, almost everywhere, fill whatever gaps are left in ensuring their households' reproduction and well-being.

In Howa most of the work associated with the political-economic aspect of social reproduction was accomplished within the household or the extended family. Seeds of change were evident, however. For instance, the role of formal schooling increased dramatically during the course of my study, decentering the household as a source of work-related knowledge. Less dramatic, but quite significant, the social extension program associated with the agricultural project introduced a nursery school, encouraged formal schooling, and sponsored a women's literacy class, among other activities that began to shift some of the weight of social reproduction from the household to the state.

There is a blurred boundary between these practices that I have associated with the political-economic aspect of social reproduction, and those associated with its cultural aspect. Here I include the cultural forms and practices associated with knowledge acquisition broadly understood, not just in relation to work or the workplace. Also included is the learning associated with becoming a member of particular social groups. All people are, of course, members of multiple and overlapping social groups, and social reproduction entails acquiring and assimilating the shared knowledge, values, and practices of the groups to which one belongs by birth or choice. Through these material social practices, social actors become members of a culture and construct their identities. In these activities, young people (and others) are both objects and agents, acquiring cultural knowledge and reworking it through the practices—intentional and otherwise—of their everyday lives. Here again, household forms and their fluid gendered and generational divisions of labor have as much bearing on how cultural reproduction is enacted as on its contours and what it is socially understood to encompass. These relations are both the "medium and the message" of social reproduction, and thus their particular form is of important political-economic and sociocultural consequence.

Other arenas of social reproduction that are primarily cultural include that broad category of cultural production categorized as "the media," "mass culture," and those institutions associated with religious affiliation and practice. "Culture" is made within these broad arenas, and in this interchange, the social relations of production and reproduction that characterize a particular social formation at a given historical moment and geographical location are encountered, reproduced, altered, and resisted. For instance, the spiritual practices associated with Islam, especially those of Sufism, were of particular significance in Howa. Islamic practice was so much a part of everyday life that children appeared to absorb it by osmosis—virtually all adults prayed five times a day, there was no alcohol available in Howa, not even the homemade beer that was widely available in many similar villages, Sufi paths and the sheikhs associated with them were a common topic of conversation, and people routinely visited religious sheikhs in and outside of the village for counsel and treatment of all manner of problems. Children played at praying, climbed rubble heaps or whatever height they could muster to bellow the call to prayer, sang ballads about religious heroes and leaders as readily as they sang popular songs, and were often entranced by the exchanges among their elders and various religious sheikhs. Important as these practices associated with the cultural realm of social reproduction were, I paid much less attention to them than to the material practices associated with work. Ironically, or perhaps not, the cultural practices associated with religion may have proven a more fruitful arena in the search for resistance than the ones I pursued, especially as Howa grappled with the differences between the familiar Sufi paths and the more rigid version of Islam propounded by the state beginning in the mid-1980s.

Finally, apart from these cultural arenas, there are the material grounds of reproduction—its environmental aspect. All modes of production produce and are enabled by particular political ecologies. This fact is so obvious that it often goes unremarked, but the environmental toll of centuries of capitalist production, and its increasingly global nature, has been enormous. The widespread and serious environmental problems that are symptomatic of capitalist relations of production have received plenty of public attention, but not necessarily as problems of social reproduction (although see O'Connor 1994). In Howa, these issues of political ecology could be seen quite directly in the landscape produced by the agricultural project, which, among other things, required chemical inputs and reduced available grazing and wooded areas, all of which compromised ongoing production. Environmental degradation—in this case part and parcel of "development"—undermines sustained productivity. In Howa there was a particular and direct toll of degradation upon children, who frequently worked in the fields. The health of children in Howa was compromised by their exposure to pesticides that were sprayed in the fields without warning while they and others were working; by their direct contact with herbicides that often covered the "weeds" they picked as greens for a family meal; and by their vulnerability to schistosomiasis from wading into the irrigation canals where young boys swam and girls and boys fetched water on days when the village well was closed.

In the course of "development" the relationship between production and social reproduction is thrown open and reconfigured. Along the way, the material social practices and cultural forms associated with social reproduction are altered. So, too, is the balance among settings where social reproduction takes place. These processes are contested and uneven, but looking closely at them should render clearer what is at stake in contemporary "development." Refracting the social development of the children through the "development" imposed on their community, and vice versa, reveals the promise and perils of the "recent future." The recent future, a time-space of becoming and immanence, is produced in the routine practices of everyday life, such as children's work and play, as much as by larger social and political-economic forces, such as those associated with global capitalism. I try to keep both in tension in this work because it seems to me that if we are ever to begin to undo the operations of capitalism, global and otherwise, we will have to understand how it roosts in the routine, and how it does not.

The focus of my project is children's work, play, and learning, and the content and organization of their knowledge about the physical environment, including knowledge of agriculture, animal husbandry, botanical and animal resources, land use practices, and environmental degradation. I emphasize that body of knowledge concerned most directly with sustained productive use of the local environment. Environmental knowledge is not only fundamental to enduring production in Howa, but with the shift from the largely subsistence dryland cultivation of sorghum and sesame to the irrigated cultivation of cotton and groundnuts for the world market brought about by inclusion in the Suki Project, it had became a fulcrum of struggle. Not only was the traditional substance of environmental knowledge threatened with obsolescence or erasure, but the means of its acquisition started to butt up against and jostle with other forms of learning. The ensuing shifts and struggles altered children's daily lives as producers in their community and their prospects as adults. I hope to render specific the abstract forms and practices associated with the production and exchange of knowledge and tie them to the particular historical, geographical, and socioeconomic conditions of Howa village as a means of both grounding my analysis of the social, cultural-ecologic, and political-economic change underway there and providing a framework to clarify the meaning and significance of everyday practices amid such transformations.

The heterogeneity and vitality of cultural forms and practices point to the indeterminate nature of social transformation. If capitalism roosts in the routine, so do the possibilities for its disruption; the chance for diversion speckles well-worn paths. These may come from any quarter. The elaborate and embedded nature of the children's everyday practices, only hinted at in the first part of this chapter, makes clear that the outcomes of externally imposed socioeconomic change are not determined, however powerful they may be, but rather are forged in the dynamic and often contradictory juncture of social, cultural, political, and economic practices composing production and social reproduction.

Capitalism is often depicted as a deluge that sweeps away and drowns all that comes before it. Yet it is, of course, more complicated than that. As Stephen Jay Gould (1982) pointed out years ago, Darwin was drawn to study earthworms because their mundane and incremental actions transform the world. As I began to see in this study, capitalism works more like Darwin's worms than a deluge. But it doesn't have the earth to itself. People encounter, oppose, and absorb the transformative effects of capitalism in the course of producing their identities, doing their work, playing, imagining themselves, constructing alliances, and carrying out their everyday lives. The incremental dislocations, reinventions, and removals inherent in such daily routines reveal—and create—a world of difference.

CONTEXT FOR READING

Pages 23–31 are omitted from this reading. Here's what you need to know...

This excerpt opens mid-argument, so here is the ground it stands on. Howa is a village in central-eastern Sudan, settled in the late nineteenth century by people who had been pastoralists. For most of its first century, families there farmed sorghum and sesame on rain-fed plots they held by custom, kept livestock, and gathered wood from nearby forests: an economy built around subsistence rather than sale, with little need for cash. That world changed in 1971, when the state folded Howa into the Suki Agricultural Project, a donor-funded scheme that turned independent farmers into tenants required to grow cotton and groundnuts for export. Katz reads this shift through two terms worth holding onto: the move from producing goods for a household's own use to producing crops for the market, and the resulting split between "production" (work geared to exchange) and "social reproduction," the daily, largely unpaid work of sustaining a household and raising the next generation. When the excerpt names "this transformation" and its "deskilling," it means exactly this upheaval.

to the remaking of geographies that disable or enable certain kinds of work—are a central focus of my project. Neither the direction nor the pace of such transformations are determined. These changes are grounded in a particular historical and geographical context, and evolve in the relation between production and reproduction as material social practices. My project deals particularly with those aspects of this relation that are associated with the production, exchange, and use of environmental knowledge, and by extension with the dislocation or reformulation of the household as a center of production and reproduction.

Three aspects of this transformation are of particular relevance to my project and will be addressed throughout the work. The first, referred to above, is the decline in the ability to produce or procure household subsistence through historic or conventional means. This problem was becoming apparent in many Howa households within a decade of its inclusion in the Suki Project and was partially attributable to the sundering of production and reproduction in the household, which was fundamental to the project's advance. The dislocation of the household as a principal site wherein production and reproduction are joined, and the deterioration of households' ability to procure the means of existence through traditionally available channels, have a direct effect on the nature and allocation of household labor time, and therefore change the role of children as actors and objects in the material social practices associated with production and reproduction. Second, the shifts in household social relations associated with the transformation of Howa can be understood schematically as a renegotiation of the relationship between patriarchal relations of production and reproduction and those of an evolving rural capitalism. The particularities and reverberations of this unfolding affected children's everyday lives and prospects in obvious and unanticipated ways. Finally, the transformation underway in Howa, which was changing the role of children as it altered the nature and content of their socialization—including their acquisition and use of environmental knowledge and its associated skills—appeared to be leading to a disharmony between what children knew and the skills and knowledge that were likely to be useful in their adulthoods. This disharmony was not only of importance for the children themselves, but in the broadest sense, had serious implications for the stability of Sudan as a whole. These three themes—the dislocation of the household as the center of production and reproduction, the changes this dislocation engendered within the household and beyond, and the disharmony between what children learned and what they were likely to need to know as adults—propelled and undergirded my inquiry. Because these transformations were set entrain most decisively by the Suki Project, I want to spend the rest of this chapter looking at how the political-ecological and political-economic terrain of Howa were altered by the project.

New Political Ecologies

The Suki Agricultural Project altered the fortunes of Howa, literally changing the grounds of everyday life as it accelerated political-economic and cultural shifts that were already underway. With the project, the subsistence-oriented economy of the village was undermined; money increasingly began to define social life and local land-use patterns were transformed. This change was marked directly on the landscape when the major canal of the project (running northerly from the Blue Nile at Suki town, forty-four kilometers, to Wad Tuk Tuk near Howa) was dug and created a fixed boundary between the people in the village and their (newly created) fields.[4] Practically, incorporation in the Suki Project meant that almost all of the dryland fields and much of the pasture and wooded land in the vicinity of the village were cleared, graded, and divided for the irrigated cultivation of cotton and groundnuts for the world market. This change all but precluded cultivation of the staple food crop, sorghum, severely reduced wooded areas nearby, and sharply curtailed the grazing land available to the small herds of goats, sheep, and cows held by villagers. Independent cultivators became farm tenants, no longer producing grain for household consumption but cultivating cash crops for international exchange under conditions set by the Ministry of Agriculture through the management of the Suki Project.

As the project was established, 250 tenancies of approximately 4.2 hectares (10 feddans) were distributed to residents of Howa (mostly male household heads) for cultivation beginning in 1971. Until 1981, when tenants lobbied for the right to add sorghum to the rotation, each tenancy was divided in two: half cotton and half groundnuts planted in a system of 100 percent rotation each year. The Ministry of Agriculture, through the project administration, set the schedules, standards, and accounting criteria for the project. It determined what crops would be grown and which cultivation practices and tools were appropriate. The Suki Project pioneered the use of an individual account system that effectively put all the economic risk on the tenant with few prospects for gain. With this system, all of the expenses associated with an individual tenancy, e.g., for seed, fertilizers, herbicides, pesticides, water, and mechanized plowing, were deducted automatically from the tenant's gross receipts from cotton. In other agricultural projects in Sudan, the receipts and expenses were historically calculated for the whole project or project subdivision, with each tenant getting an even share of the net receipts (see Barnett 1977; Bernal 1991).

The "individual account system" resonates with moves toward "Post-Fordism" or flexible specialization taking place in industrialized countries at the same time. Project administrators argued that the system would provide greater incentives for each tenant. However, it proved to be so difficult to derive any income at all from cotton that few tenants produced at a level where incentives might have come into play. In any case, some years tenants were not paid even what little they were owed by the project, and poorer tenants were forced to borrow money just to acquire the supplies necessary to begin the next season (see Hummaida 1977, who reports tenant distrust of the Suki management after it failed to pay farmers following the 1973, 1974, and 1975 seasons). Likewise, even though the labor demands of the harvest were high, and most tenants relied on hired assistance, they had difficulty paying the field workers without advances from the Suki Agricultural Corporation, which were often slow in coming. The empty-handed project manager's response was to encourage the use of family labor power (Y. Mahmoud 1977).

The project largely supplanted sorghum cultivation. In theory, tenant earnings were to have been sufficient to allow for the purchase of foodstuffs grown more efficiently elsewhere in Sudan (notably the rapidly expanding mechanized grain-growing areas to the northeast). In practice, tenant income was frequently inadequate for that purpose, particularly in the face of often drastic increases in sorghum prices after 1971. Not only had the vulnerability of each farm family increased in the absence of their traditional grain stores, but Sudan as a whole began to experience food deficits in the 1980s, despite its productive capacity and the dramatic expansion of mechanized sorghum cultivation in eastern Sudan. These periods of food insecurity, and even famine, did not stop the politically powerful mechanized farmers from exporting their harvests to Saudi Arabia and the Gulf States, often as camel fodder. Their unseemly business decisions, unfettered by any government restriction, contributed to increasing food insecurity and fueled domestic price increases for sorghum.

During the 1970s, for instance, the price of sorghum rose from the equivalent of US$2.50 per sack (about 95 kilograms) to as high as US$50.00 per sack. The average household used upward of twenty sacks of sorghum each year, so this increase was an assault on household budgets across Sudan. The price increase was particularly difficult for the former subsistence cultivators of Howa, because their burden was double-edged; not only were they forced to purchase sorghum at prices inflated by as much as 2,000 percent, but they were frustrated that they were no longer able to grow sorghum, the surplus of which they might have sold at these astronomical prices.

Because of the difficulties and frustrations presented by the price increases, the Suki Farm Tenants Union agitated for, and in 1981 received, permission to cultivate sorghum in their tenancies at their own discretion. The twist was that they were only allowed to do so on one hectare of the portion of their fields allocated to groundnuts. Although groundnut prices were quite high that year, most tenants opted to replace half of their crop with sorghum as soon as it was permitted by the Ministry of Agriculture beginning in the 1981–82 season. Their eagerness testifies to the difficulties most tenant households faced in procuring subsistence under the conditions of production associated with the Suki Agricultural Project, even in years when their cash crops fetched lucrative prices. Many tenants, however, frustrated with the difficulties of cultivating cotton and the lack of earnings from it, wanted to change the Suki crop rotation to include wheat and possibly other food crops in addition to the sorghum and groundnuts, and if necessary, cotton. They were not successful in these efforts.

If, in the first few years of the project, most tenants were able to garner an income from cotton, few were able to do so by the close of its first decade in 1981. The reasons for this are threefold. First, the adjusted world price of cotton had declined since the project's first year of operation in 1971. Second, the price paid to tenants was established each year by the government marketing board below world-market rates. Third, all expenses associated with the tenancy, including those for groundnuts, were

Table 1. Economics of Cotton Cultivation, Suki Agricultural Project, 1980/81

Average income from cotton production

Average price per qantar = $38.70

Average yield = 2.156 qantar/feddan

Average total output on 5 feddans = 10.78 qantars

Average total gross income = $417.19

Costs of production

Per feddan	
Land and water for cotton	$20.00
Land and water for groundnuts	15.00
Pesticide spraying	48.71
Urea	26.02
Marking materials	2.61
Sacks for cotton	1.41
Total (Deducted per feddan of cotton cultivated)	$113.75
Total deductions on 5 feddans	$568.75
Total deductions excluding cost of groundnut cultivation	$493.75
Per qantar	
Sacks for seeds	$ 1.02
Ginning	7.98
Transport to ginning	.98
Interest on loans	3.13
Social services and auditing fees	1.79
Herbicide, extension services, rat protection, cotton classification, guarding, and employee bonuses	1.55
Total (Deducted per qantar of cotton produced)	$16.45
Total costs on average yield (10.78 qantars)	$177.33

Total income

Average total costs (producing 10.78 qantars of cotton on 5 feddans) = $746.08

Average net income/(loss): $746.08 deducted from $417.19 = ($328.89)

Average net income/(loss), excluding costs of groundnut cultivation = ($252.33)

Note: All amounts given in U.S. dollars, 1981 exchange rates. 1 feddan = 1.038 acres. 1 qantar = 100 pounds of ginned or 315 pounds of unginned cotton. All costs of groundnut production are deducted from cotton accounts.

Source: Figures from the Offices of the Suki Agricultural Corporation in Wad Tuk Tuk given to the author by the assistant director of the project, Abdel Rahman Mohamed Abdel Rahman. Price figures from the *OfW-cial Prices Paid by the Cotton Public Corporation,* obtained from its offices in Khartoum, Sudan.

deducted from the cotton account and these had risen as cotton prices had remained stagnant. By 1981 production expenses were so high that a tenant had to produce 5.11 qantars of cotton per feddan (547 kilograms ginned cotton per hectare) simply to break even (see Table 1). The average yield across the Suki Project was less than half that. The exploitation concealed in these figures becomes clear with reference to the original feasibility study for the project, conducted by Hunting Technical Services in the early 1960s, which estimated yields of 4.5 qantars per feddan *under good conditions* (MacDonald et al. 1964, 92). Conditions had not been "good." From 1971–81 the average yield exceeded 4.5 qantars only twice. The figures in Table 2 make clear the difficulty tenants faced in earning income from cotton. Even under the well-funded and intensively supervised colonial administration of the Gezira Project, the average cotton yield between 1912 and 1946 was 3.78 qantars per feddan, and only exceeded 5 qantars per feddan in three of these thirty-five years (Tothill 1948, 786). The Suki tenants were impoverished by the very premises of the "development" project in which they were enmeshed.

Reflecting on these economic relationships, many tenants volunteered that they put relatively little effort into their cotton crop in comparison to their groundnuts. Few tenants expressed concern, for instance, if they learned that their cotton field was damaged or destroyed by grazing animals, crop pests, or irrigation problems. The contrast with the level of attention, effort, and concern for groundnut cultivation was marked. Likewise, many spent time growing sorghum on the margins of the agricultural project or in their old bildat fields when these had not been incorporated in project lands (Hummaida 1977). For these reasons, among others, cotton yields in the Suki Project declined steadily from a high of 6.65 qantars per feddan in the 1976–77 season to an all-time low of 2.156 in 1980–81 (see Table 2). Subsequent years have been similarly dismal. Throughout Sudan, however, cotton yields have declined since the 1970s because of lowered world prices, a lack of supplies and spare parts, the increased costs of inputs following oil shocks, labor shortages, acts of neglect, and infrastructural difficulties. In contrast, there was no decrease in the average yield of groundnuts in the Suki area between 1971 and 1981 (see Table 3).

One factor contributing to low crop yields in the Suki Project was the production schedule established for cultivating cotton and groundnuts. The timetables for sowing, weeding, and harvesting both crops were virtually identical. The overlap had been predicated on groundnut cultivation being highly mechanized. However, almost since the project's inception, breakdowns of machinery worsened by a severe lack of spare parts thwarted the mechanization of groundnut cultivation. Moreover, most farmers could not easily afford or were reluctant to pay the fees charged for using available equipment; they preferred to do the work themselves with the assistance of household members or hired day laborers.

The failure to mechanize groundnut farming, along with the increased attention given to the crop by tenants who stood to make substantial private gains from its sale, drained attention from the cotton crop. This imbalance was likely to be further exacerbated in years when the world price for groundnuts was inflated. Such was the case,

Table 2. Cotton Cultivation in the Suki Agricultural Project, 1971/72–1980/81

	1971/72	1972/73	1973/74	1974/75	1975/76	1976/77	1977/78	1978/79	1979/80	1980/81
Total number of tenants	7,345	7,174	7,268	7,523	7,847	7,782	7,771	8,000	7,700	—
Total number of active tenants	7,345	7,143	7,232	7,434	7,689	7,702	6,346	6,228	7,700	—
Total number of productive tenants	5,942	6,703	6,691	6,935	7,366	7,702	6,345	6,228	6,226	—
Total area (in feddans)	36,730	35,870	36,340	37,615	39,237	38,910	38,855	40,000	38,500	—
Prepared area (in feddans)	36,730	35,715	36,160	37,170	38,443	38,510	38,800	37,842	38,500	33,930
Harvested area (in feddans)	29,712	33,515	33,455	34,675	36,831	38,510	31,730	31,140	31,130	27,670
Acreage yield (qantars/feddans)	2.9	3.5	4.1	3.78	4.01	6.65	6.02	4.01	2.63	2.156

Source: Records of the Suki Agricultural Corporation in Wad Tuk Tuk, Sudan.

Table 3. Groundnut Cultivation in the Suki Agricultural Project, 1971/72–1980/81

	1971/72	1972/73	1973/74	1974/75	1975/76	1976/77	1977/78	1978/79	1979/80	1980/81
Total number of tenants	6,193	7,177	7,325	5,723	7,413	5,707	5,887	7,374	6,311	—
Total number of active tenants	3,940	4,776	3,265	4,391	4,702	4,414	4,633	5,046	5,964	—
Total area (in feddans)	34,565	35,885	36,626	28,615	37,065	28,536	23,167	36,870	31,555	29,580
Total productive area (in feddans)	19,700	23,730	16,325	21,955	23,510	22,070	29,435	25,230	29,820	15,093
Average yield (100 pound sacks/feddan)	4.13	6	6.03	10	24	26.5	25	26	30	24

Source: Records of the Suki Agricultural Corporation in Wad Tuk Tuk, Sudan.

for example, in 1980 and 1981, when the price of groundnuts had risen fivefold from US$2.50 to $12.50 a sack. This high price, combined with depressed cotton prices and the general deterioration of project equipment and infrastructure, helps to explain the steady decline in cotton yields and the relative stability of groundnut yields across the Suki Project from 1976 to 1981, when despite its lucrativeness, groundnut cultivation was sacrificed for sorghum.

Regardless of productivity levels, the labor requirements for both cotton and groundnut cultivation were significant. As many tenants pointedly noted, they were substantially higher than those associated with the dryland cultivation of sorghum and sesame. Most tenants relied on a mix of labor from family members and hired hands. With almost all tenants being men, the increased labor demands of cultivating in the Suki Project conscripted wives', children's, and other relatives' labor time so that tenant households could stay afloat. My research suggested that the Suki Project drew children from school to the fields more intensely than farming before the project had. While access to land and other productive resources in Howa was mediated by men's participation in various social networks—however changed these networks might be as a result of the Suki Project—their access to labor was individualized, and the project was predicated on tenants mobilizing household and family labor (see Berry 1993, 17). Indeed, preference was given to larger families when tenancies were first allocated, and during labor crunches, when the project management was unable to advance tenants money to pay hired hands, they explicitly encouraged the conscription of family labor (Hummaida 1977; Y. Mahmoud 1977).

The major agricultural tasks were clearing, planting, thinning, weeding, irrigating, and harvesting. The agricultural season began in late May with clearing the fields. Crops were sown in June and early July and from then until November there was almost constant work as the plants were thinned, weeded, and irrigated. The groundnut harvest began in early November, and about six weeks later cotton was ready to be picked. By the end of January the bulk of the year's agricultural work was finished. Tenants who had not yet marketed their crops generally did so in February. Following the agricultural season was a time for celebration. Marriages and circumcisions were commonly celebrated in Howa during March and April because both time and money were relatively plentiful.

Rarely did a tenant finish an agricultural task without the assistance of household members. Their efforts were supplemented in some instances by communal work parties (*nefir*) or hired laborers. While wealthier tenants could afford to hire either day laborers from the village or seasonal laborers from elsewhere in Sudan (usually the south or far west), most tenants kept hired labor to a minimum, and relied instead on family members, except during the height of the labor-intensive harvest period, when work was accomplished almost entirely by village women and children, especially girls, working for hire or in the fields of their own households.

For the average tenant, then, most agricultural tasks were completed by family members. Fields were generally cleared by the tenant, sometimes with the help of teenaged sons or other adult male relatives. During the planting period, children between eight and thirteen years old provided most of the assistance to their fathers (and occasionally grandfathers or uncles without their own children to assist them). Thinning, weeding, and irrigating were completed by tenants themselves, often with the help of adult male relatives and sometimes with communal work parties or hired labor. Of these tasks, weeding was the most labor-intensive, and young boys frequently assisted their fathers and uncles. The communal work parties (*nefir*), which had long been a mainstay of Sudanese agricultural life, had become labor exchanges by the time of the Suki Project. A group of men (often relatives) would work for one tenant or another in exchange for a return of the favor and possibly a meal. In this way, people pooled their resources during labor-demanding periods apart from the harvest. Since the harvest was accomplished largely by local women and girls or migrant laborers, and women's earnings were considered separate from household (and by extension tenancy) earnings, their labor did not exchange in the same way that male tenants' did.

The harvest was the most labor-demanding period of the agricultural year. In many tenant households, the whole family worked from dawn until sunset, breaking only during the hottest part of the day. Some families even built temporary shelters (*cambo*) near their fields to maximize their labor time. In the nearby Rahad Project, estimates were that one laborer was necessary to pick each feddan of cotton (Jansen and Koch 1982). Yet, as Jansen and Koch note, after the establishment of the Rahad Project in 1978, the number of agricultural laborers who turned up was hardly more than half of what was required in any given year. Such localized labor shortages have been common in many parts of rural Africa, and are exacerbated as more people leave the countryside in search of more permanent or lucrative employment (cf. Berry 1993). Likewise in Howa, where, according to Jansen and Koch's figures, the equivalent of about 1,250 cotton pickers would be required each season. Nowhere near that number of migrant workers was available, and the harvest was routinely accomplished by local women and girls working intensively from tenancy to tenancy.

Traditional agriculture had not given way completely to the irrigation project in Howa, although the total time and space devoted to it was small compared with irrigated cultivation. With most of the arable land surrounding the village incorporated into the project, little land was available for rain-fed cultivation. Some people still managed to have bildat (dryland fields) where they grew sorghum and occasionally millet. When the cultivation of sorghum was permitted in the irrigated fields in 1981, the area devoted to rain-fed sorghum cultivation was reduced further, although there is evidence that this made room for the renewed cultivation of sesame, albeit on an extremely limited scale. Others grew a few vegetables, either in *juruf* on the riverbanks or along the margins of their irrigated fields. Given the steepness of the Dinder's banks, just one or two old-timers attempted to cultivate horticultural crops there. A few more grew vegetables and sweet sorghum (*Sorghum spp. graminae*) in the margins of their tenancies or (surprisingly rarely) in kitchen gardens, where loofa gourds often sprawled exuberantly in the rainy season. Most of these crops were for household consumption.